

AJAY V A

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PROFESSIONAL SUMMARY

Python Developer with solid experience in backend development and a strong passion for AI, machine learning, and data-driven problem-solving. Proficient in **Python**, **Django**, **Frappe**, and **REST API** development, with practical experience in **Machine Learning**, **Deep Learning**, **Computer Vision**, and **NLP**. Skilled in converting business needs into efficient backend workflows, optimizing APIs, and building scalable automation features, while delivering clean, reliable code through strong problem-solving, collaboration, and continuous learning. Passionate about applying AI to real-world challenges and building impactful, high-performance systems.

WORK EXPERIENCE

Python Developer — Faircode Infotech

Oct 2024– Jan 2026

- Developed and optimized **REST APIs** for the Ship Design Approval System, enabling reliable data exchange and smoother departmental workflows.
- Implemented multi-stage approval pipelines with detailed status tracking, improving transparency and efficiency throughout the pre-construction review process.
- Developed real-time analytics dashboards using **Matplotlib** for visual charts and performance insights.
- Built automated report generation modules using python-docx and **Pandas**, creating structured tables, formatted documents, and data summaries for stakeholders.
- Collaborated closely with clients and project leads to interpret business requirements, propose technical solutions, and refine backend modules for usability and performance.
- Enhanced backend performance by optimizing **Frappe ORM** queries, improving API response times, and ensuring clean, maintainable, and scalable code.

Machine Learning & AI Intern — Gofreelab Technologies

Jun 2024– Sept 2024

- Developed Machine Learning and Deep Learning solutions across Computer Vision and NLP domains using real-world datasets.
- Built face classification systems covering preprocessing, embedding extraction, model inference, and deployment.
- Applied model evaluation, dataset curation, and modular pipeline design to ensure scalable, reliable, production-ready ML workflows.

PROJECTS

SHIP BUILDING DIGITALIZATION PLATFORM

- Designed and developed a scalable backend system powering the ship design approval lifecycle, with optimized ORM operations and secure workflow handling.
- Implemented structured role-based workflows with multi-stage validations, automated notifications, and standardized reporting pipelines.
- Built real-time analytics dashboards with data aggregation, approval metrics, and department-wise performance insights.

FACE AUTHENTICATION SYSTEM

- Developed a real-time face authentication system using **FaceNet** and **MTCNN** for face detection and embedding generation, supporting both **recognition (1:N)** and **verification (1:1)** workflows.
- Optimized similarity search by integrating **FAISS**, replacing brute-force matching to improve scalability and performance for larger datasets.
- Designed a modular ML pipeline with separate components for detection, embedding, and matching, ensuring clean architecture and reusability.
- Built an interactive Streamlit UI with real-time webcam verification and implemented a logging system to track authentication events with timestamps and performance metrics.

MULTIMODAL DOCUMENT CLASSIFICATION SYSTEM

- Built a multimodal document classification system using **ResNet** and **DistilBERT**, achieving ~92% accuracy across diverse document types.
- Integrated OCR using **EasyOCR** with custom text preprocessing to extract and clean textual features from document images.
- Designed a late fusion pipeline combining visual and textual predictions, and optimized performance via OCR caching and modular architecture.
- Developed a Streamlit app with real-time classification, confidence scoring, and automated document routing into category-wise folders.

DRIVER DROWSINESS DETECTION SYSTEM

- Developed a safety system using **Python** and **Arduino** to detect driver fatigue.
- Applied **computer vision** and ML models to track eye movement and trigger real-time alerts.
- Improved accuracy and reduced false alarms through optimized image processing.

SKILLS

Languages: Python , SQL , C#

Frameworks & Libraries : Django, Frappe, Flask, scikit-learn, OpenCV, Pandas, NumPy, Matplotlib

Databases : MySQL, SQLite , SQL Server

Tools : Gitlab, Postman, VS Code , Jupyter ,Pycharm

Concepts : Machine Learning, Deep Learning, Computer Vision, NLP, RESTful APIs, Data Processing

EDUCATION

B.Tech in Computer Science Engineering

APJ Abdul Kalam Technological University.

Thejus Engineering College, Thrissur, India

2019 – 2023

CERTIFICATIONS & LEARNING

Microsoft Certified: Azure AI Fundamentals (AI-900)
(2025)

Python and Data Science with AI — ExpertzLab Technologies
(2023 - 2024)